Lekai Qian

Email: lekaiqian@outlook.com | Website: lekai-qian.github.io | GitHub: github.com/Lekai-Qian

Education

South China University of Technology (SCUT), Guangzhou, China *Sep 2023–Jun 2027 (expected)*
Undergraduate Student in Artificial Intelligence, School of Future Technology
— GPA: 3.76/4.00; TOEFL iBT: 100

Research Interests

My research centers on **generative modeling**, aiming to improve quality, structural coherence, and controllability. I study how representations, architectures, and conditioning shape generative behavior across modalities. My current work uses symbolic music as a testbed, beginning with representation and tokenization design.

Research Experience

MBZUAI, Abu Dhabi, UAE (Remote) *Jun 2025–Jun 2026*
Undergraduate Researcher, Music X Lab
— Advisor: Prof. Gus G. Xia

- Led *BEAT* end to end, including representation design, large-scale data processing, model training, and evaluation; resulted in a first-author ICML 2026 paper.
- Collaborated with Prof. Xiaosong Ma's team on *StreamMUSE*, designing and training the model adopted for real-time accompaniment; acknowledged in the RTAS 2026 paper.

SCUT, Guangzhou, China *Feb 2025–Feb 2026*
Project Lead, Undergraduate Research Program
— Project: Cascaded Diffusion for Music Generation
— Advisor: Prof. Qi Liu

- Led an eight-member team to build a 200K-piece dataset and a two-stage cascaded diffusion framework with compressed latent representations; received an *Excellent* final evaluation.

Publications

L. Qian, H. Gu, J. Zhao, and Z. Wang, "BEAT: Tokenizing and Generating Symbolic Music by Uniform Temporal Steps," in *Proceedings of the 43rd International Conference on Machine Learning (ICML)*. Accepted. 2026.
Contribution: Introduces a time-centric representation that makes musical time explicit and improves sequence modeling. [Paper] [Demo]

L. Qian*, H. Gu*, D. Li*, B. Cao, and Q. Liu, "Pianoroll-Event: A Novel Score Representation for Symbolic Music," in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. Accepted. 2026.
Contribution: Compresses sparse piano-roll structure into efficient event sequences while preserving musical patterns. [Paper]

H. Gu, **L. Qian**, H. Zhou, Q. Liu, and S. Wang, "BeatEdit: Symbolic Music Generation as Explicit Editing," in *Proceedings of the 34th ACM International Conference on Multimedia (ACM MM)*. Accepted. 2026.
Contribution: Frames generation as explicit editing for controllable correction, completion, and accompaniment. [Paper]

B. Cao*, **L. Qian***, D. Li, H. Gu, M. Xu, and Q. Liu, "Anchored Cyclic Generation: A Novel Paradigm for Long-Sequence Symbolic Music Generation," in *Findings of the Association for Computational Linguistics (ACL)*. Accepted. 2026.
Contribution: Improves long-range structural coherence through an anchored cyclic generation paradigm. [Paper]

*Equal contribution.

Invited Talk

KSMI Post-ICML Workshop, Seoul, South Korea *Jul 13, 2026*
Invited Talk: "BEAT: Tokenizing and Generating Symbolic Music by Uniform Temporal Steps"
Invited by Prof. Juhan Nam, KAIST